

Derick Aaron Lievian

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Indonesian Citizen

Education

The Chinese University of Hong Kong, Shenzhen | Shenzhen, China

Expected: May 2029

School of Data Science, Bachelor of Science in Computer Science and Engineering

GPA: 3.756/4.000

Half Entry Scholarship Recipient

Relevant Coursework

Calculus 1 (MAT1001), Calculus 2 (MAT1002), Linear Algebra and its Applications (MAT2041), Introduction to Data Science (DDA2001), Introduction to Computer Science and Java Programming (CSC1003), Computational Laboratory using Java (CSC1004)

Technical Skills

- **Languages:** Java, JavaScript, TypeScript, HTML, CSS, Python, C#
- **Frameworks/Utilities:** Node.js, Express.js, Next.js, React.js, MongoDB, Tailwind CSS, JavaFX, Supabase, Canvas.js
- **Tools:** Git, GitHub, Bash, PowerShell, REST APIs

Projects

- **AI Learning Toolkit** | Next.js, Gemini API, Langchain, MongoDB, Supabase, Pinecone, Tailwind CSS

Architected a full-stack Retrieval-Augmented Generation (RAG) platform designed to streamline academic study workflows. Developed an automated pipeline that incorporates the Gemini API to create flashcards from the uploaded PDFs. Engineered a Q&A interface using LangChain and Pinecone vector embeddings, enabling users to ask questions on their documents. Managed a hybrid data architecture utilizing MongoDB for user metadata, Supabase for cloud object storage, and secure JWT-based authentication.

- **Chat Room** | Java, Socket Programming, Multithreading

Engineered a real-time communication server supporting multiple concurrent clients using ClientHandler and synchronization. Implemented robust socket I/O and thread-safe broadcasting to ensure message integrity and server stability.

- **Automated YouTube Shorts Generator** | Python, MoviePy, YouTube Data API, REST APIs

Engineered an end-to-end Python pipeline that automates the creation and deployment of "Fun Fact" YouTube Shorts, achieving a peak of 2.3k views in a video. Integrated multiple REST APIs (YouTube, Images, and Quotes) to dynamically source content, ensuring a 100% automated workflow from data collection to video rendering. Used MoviePy to overlay images and text, and structure the overall video. Utilize Google's text-to-speech for reading the caption of the fun fact quote.

Languages

- English - Professional Working Proficiency
- Bahasa Indonesia - Native Proficiency
- Mandarin Chinese - Limited Working Proficiency